



SA30 Series Fabry-Pérot Interferometers

User Guide



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Chapter 1 Safety



Warning

Take appropriate measures for eye safety.

Never look directly into the beam, not even at the output of the SA30.



Shock Warning

Never touch the output of the SA201B Fabry-Pérot Controller.

Chapter 2 Description

The SA30 is a high finesse Spectrum Analyzer used to examine the fine structures of the spectral characteristics of CW lasers. The spectrum analyzer consists of a near-confocal cavity that contains two high reflectivity mirrors. By varying the mirror separation with a piezoelectric transducer, the cavity acts as a very narrow bandpass filter. Knowing the free spectral range of the SA30 allows the time-base of an oscilloscope to be calibrated to facilitate quantitative measurements of a laser line shape. With a resolution of below 1 MHz, the fine structure resulting from multiple longitudinal modes of a laser line can be resolved.

For discussions on the spatial mode structure, transmission spectrum, and finesse of a Fabry-Pérot interferometer, please see the full web presentation at **www.thorlabs.com**.

Chapter 3 Setup

3.1 Optical Setup



Warning

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The SA30 should be mounted so that it can be easily aligned to the incoming beam. It is recommended that Thorlabs' KM200 Ø2" Kinematic Mount be used to mount the interferometer at the 2-inch diameter flange. Also, it is recommended that a fold mirror is used to direct the beam into the Fabry-Pérot interferometer. For the SA30 series, the EFL and position of the lens depend on the beam parameters before the lens and must be chosen in a way to obtain the recommended waist radius. The recommended waist radius of the incident lens is given by

$$w_0^{inc} = \frac{\lambda f}{\pi w_0^{rec}} \frac{1}{\sqrt{1 + \frac{f^2}{(z_0^{inc})^2}}} \quad (1)$$

where w_0^{inc} , w_0^{rec} , z_0^{inc} , λ , and f are the beam waist size of the beam incident on the lens, the recommended beam waist radius at the center of the instrument, the Rayleigh range of the incident beam on the lens, the wavelength of the incident light, and the focal length of the lens. For a well-collimated beam, i.e., when the Rayleigh range of the incident beam on the lens is much larger than the focal length of the lens and the beam is only weakly diverging, Equation (1) reduces to

$$w_0^{inc} = \frac{\lambda f}{\pi w_0^{rec}} \quad (2)$$

The lens should be placed at a distance D

$$D = \frac{f}{1 + \frac{f^2}{(z_0^{inc})^2}} \quad (3)$$

away from the center groove. This reduces to $D = f$ for a well collimated beam^a. Please note that the Rayleigh range of a Gaussian beam is given by

$$z_0 = \frac{\pi w_0^2}{\lambda}, \quad (4)$$

where w_0 is the beam waist radius and λ is the wavelength of the incident light.

The recommended waist radius for the SA30 series is given by

$$w_0^{rec} = \sqrt{\frac{\lambda}{2\pi} \sqrt{d(2R - d)}}, \quad (5)$$

^a P. W. Milonni and J. H. Eberly, *Lasers* (John Wiley & Sons, Inc., 2010) p. 290.

where R is the radius of curvature of the mirrors (50 mm), d is the distance between the two resonator mirrors (50 mm), and λ is the wavelength of the incident beam^b.

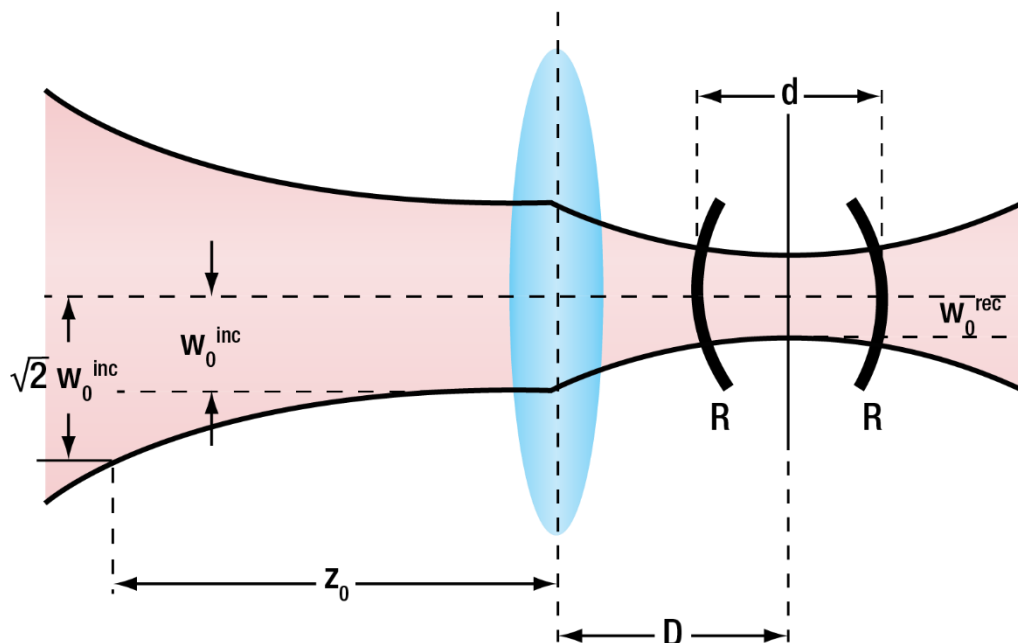


Figure 1 Schematic showing the incident beam waist and recommended beam waist.

3.2 Driving Electronics and Oscilloscope



Caution

Caution is given when the SA30 is used with another driver than the SA201B.

Please make sure to never apply voltages below 0 V or above 150 V.

In a typical application, the SA30 interferometer is used in conjunction with a signal generator and an oscilloscope, as shown in Figure 2. The signal generator must produce either a triangle or saw-tooth wave with an adjustable frequency, amplitude, rise time, and offset; please refer to the Thorlabs SA201B Fabry-Pérot controller manual for details. Please note that the Thorlabs SA201B Fabry-Pérot controller was used to generate the scan signals required for obtaining the data in this document.

The signal generator is used to repetitively scan the length of the cavity to sweep through one full spectral range (FSR) of the interferometer. An oscilloscope is typically used to view the spectrum and make quantified measurements of spectral features. The DC Offset adjustment on the SA201B is conveniently used to center a mode on the scope. If the detector is connected directly to the scope, a 5 k Ω terminator is needed.

^b H. Kogelnik and T. Li, "Laser Beams and Resonators," *Applied Optics*, vol. 5, no. 10, 1966.

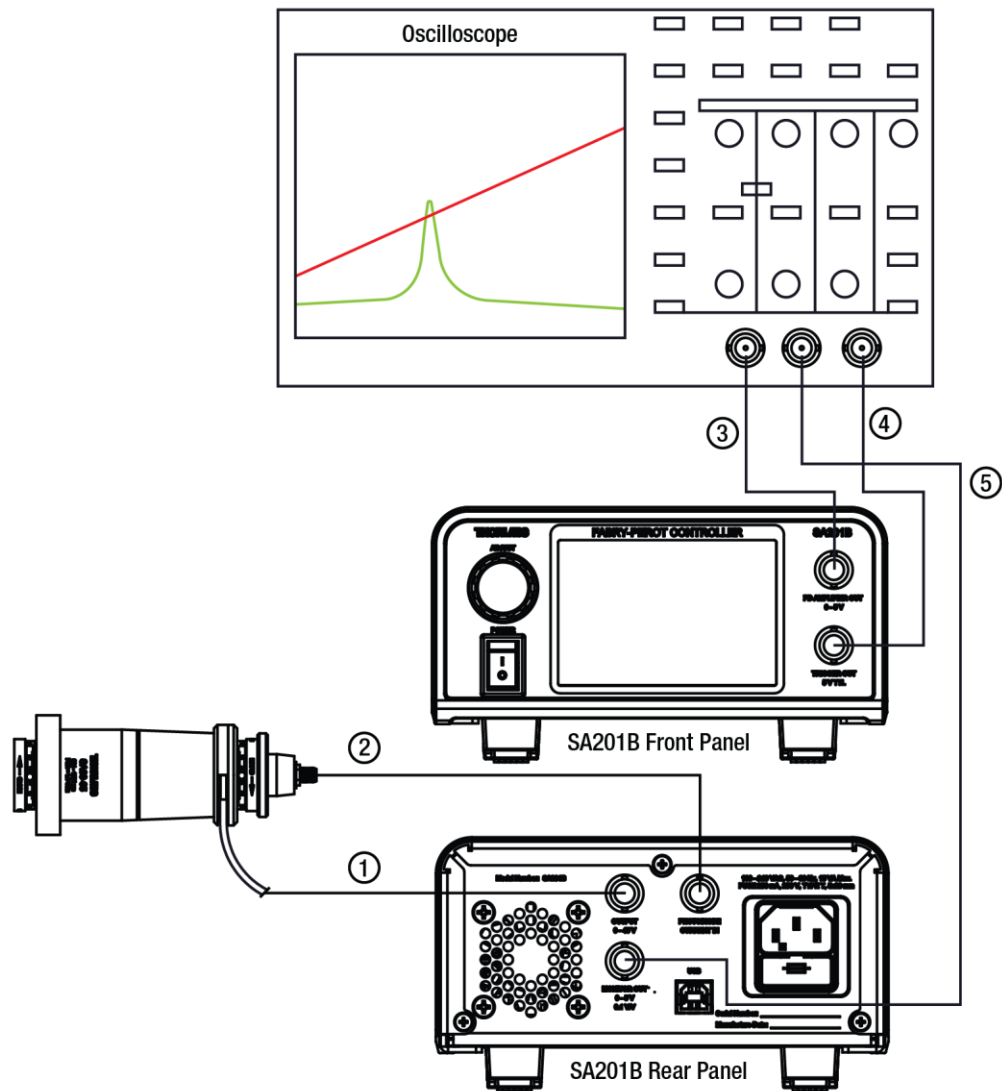


Figure 2 **Schematic of a recommended setup for using an SA30 series interferometer in combination with a SA201B controller.**

Connection	Description
1	Controller (BNC) to Piezo (Cable is Attached to FP Interferometer)
2	Photodiode (SMA) to Controller (BNC) Cable (Included with FP Interferometer)
3	Amplified Photodiode Output (BNC) to Oscilloscope Cable (Not Included)
4	Trigger Output of Controller (BNC) to Oscilloscope Cable (Not Included)
5	Monitor Out (BNC) to view the Piezoelectric Transducer signal (Not Included)

Chapter 4 Operation



Warning

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To set up the SA30 series Fabry-Pérot interferometer, mount the unit in a KM200 mirror mount and make the connections according to the description in the SA201B manual or the piezo controller and detector amplifier you are using. Then, use the iris apertures to align the incoming beam to the optical axis of the SA30. Note that the detector from the back of the unit can be removed and mounted on its own to aid in the initial alignment. To align the beam along the optical axis, close the input iris and center your beam on the iris opening. Leave the back iris completely open and start to scan the unit. Now, use the tip/tilt adjustment until the beam is centered through the body of the SA30. Adjust the scope gain to maximum sensitivity, position the detector close to the rear opening, and slowly close the back iris while simultaneously correcting the two angular adjustments on the mirror mount. The beam is close to an overlap with the optical axis when each second mode in the spectrum becomes somewhat smaller. Once the beam is centered, mount the detector back on the main body. For this alignment sequence it is convenient to have the oscilloscope triggered by the piezo voltage scan; the SA201B does provide such a trigger signal.

Compared to the models in the SA200 and SA210 series, the SA30 instruments require more sophisticated alignment. Therefore, once initial alignment has been established according to the above description, zoom in on a transmission mode and optimize the alignment for maximum suppression of higher order modes, which typically occur within a few MHz around the fundamental transmission modes. For this, it is very convenient to set the trigger on the oscilloscope to one of the larger transmission modes visible in the spectrum. Now, fine tune the alignment of the SA30 with respect to the incident beam. Also, it might be necessary to fine tune the distance between the lens and the SA30 within a few centimeters around its initial 25 cm distance described in the previous chapter. Once higher order modes are suppressed, the unit is ready for measurements. The left oscilloscope panel in the figure below shows an alignment that can be improved. Since the higher order transverse modes are still visible, the lens position or beam alignment needs to be changed. In contrast, the right panel shows a perfectly mode-matched alignment.

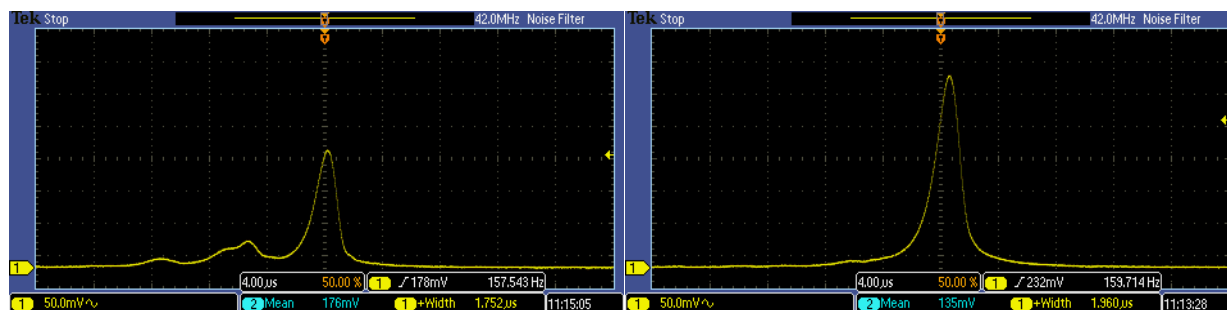


Figure 3 *Left: Higher order transverse modes are still visible and the alignment has to be improved.*
 Right: Good alignment, higher order transverse modes are not evoked.

Chapter 5 Spectrum Analyzer Controller and Accessories

- The **SA201B** controller generates a voltage ramp, which is used to scan the separation between the two cavity mirrors. A photodiode is used to monitor transmission of the cavity. Using the Trigger Out signal from the controller, an oscilloscope can be used to display the spectrum of the input laser. The controller provides adjustment of the ramp amplitude (0 to 30 V) and scan-time (10 ms to 10 s) to allow the user to choose the scan range and speed. A DC offset control (0 to 15 V) is provided to allow the spectrum displayed on the oscilloscope to be shifted right or left, and zoom capability provides up to 100X increase in spectral resolution.
- Thorlabs' **KM200** Ø2" kinematic mount can be used to mount the SA30 Scanning Fabry-Pérot Interferometer.

Chapter 6 Specifications

Laser Output Power (Maximum)	
Piezoelectric Transducer Input Voltage	0 to 150 V
Free Spectral Range ^c	1.5 GHz
Minimum Finesse	1500
Resolution	<1 MHz
Cavity Length ^d	50 mm
On-Resonance Transmission	10% to 20% (Typical)
Outer Housing Material	Black Anodized Aluminum
Fabry-Pérot Cavity Material ^e	Low Thermal Expansion Invar
Dimensions	Ø2" Flange; Total Length: 5.85"

Item #	Wavelength Range	Mirror Substrate
SA30-47	400 - 535 nm	UV Fused Silica
SA30-57	490 - 640 nm	UV Fused Silica
SA30-73	630 - 824 nm	UV Fused Silica
SA30-95	824 - 1071 nm	UV Fused Silica
SA30-120	1060 - 1350 nm	UV Fused Silica
SA30-144	1250 - 1625 nm	IR-Grade Fused Silica (Infrasil®)

Please refer to the Fabry-Pérot Interferometers product catalog page on www.thorlabs.com for a complete and updated list of available models.

^c FSR is set by the length of the near-confocal cavity and is given by $FSR=c/4R$, where R is the radius of curvature of the mirrors; in this case $R = 50$ mm.

^d Nominal distance between mirrors.

^e The thermal design balances the small coefficient of thermal expansion of the Invar body with the negative coefficient of thermal expansion of the piezo actuators.

Chapter 7 Mechanical Drawings

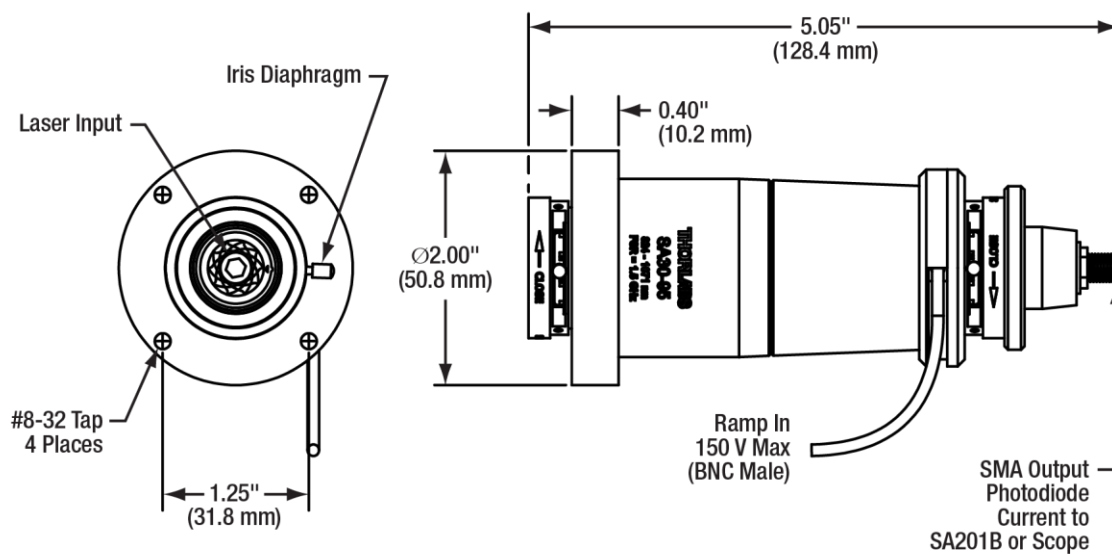
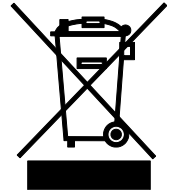


Figure 4 Mechanical Drawing for SA30-47, SA30-57, SA30-73, SA30-95, SA30-120, and SA30-144

Chapter 8 Warranty and RMA Information

Thorlabs verifies our compliance with the WEEE (Waste Electrical and Electronic Equipment) directive of the European Community and the corresponding national laws. Accordingly, all end users in the EC may return “end of life” Annex I category electrical and electronic equipment sold after August 13, 2005 to Thorlabs, without incurring disposal charges. Eligible units are marked with the crossed out “wheelie bin” logo (see right), were sold to and are currently owned by a company or institute within the EC and are not disassembled or contaminated. Contact Thorlabs for more information. Waste treatment is your own responsibility. “End of life” units must be returned to Thorlabs or handed to a company specializing in waste recovery. Do not dispose of the unit in a litter bin or at a public waste disposal site. It is the user’s responsibility to delete all private data stored on the device prior to disposal.



Annex I

8.1 Return of Devices

This precision device is only serviceable if returned and properly packed into the complete original packaging including the complete shipment plus the cardboard insert that holds the enclosed devices. If necessary, ask for replacement packaging. Refer servicing to qualified personnel.

Chapter 9 Thorlabs Worldwide Contacts

For technical support or sales inquiries, please visit us at www.thorlabs.com/contact for our most up-to-date contact information.



USA, Canada, and South America

Thorlabs, Inc.
sales@thorlabs.com
techsupport@thorlabs.com

Europe

Thorlabs GmbH
europe@thorlabs.com

France

Thorlabs SAS
sales.fr@thorlabs.com

Japan

Thorlabs Japan, Inc.
sales@thorlabs.jp

UK and Ireland

Thorlabs Ltd.
sales.uk@thorlabs.com
techsupport.uk@thorlabs.com

Scandinavia

Thorlabs Sweden AB
scandinavia@thorlabs.com

Brazil

Thorlabs Vendas de Fotônicos Ltda.
brasil@thorlabs.com

China

Thorlabs China
chinasales@thorlabs.com



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www.thorlabs.com
